

AI AGENT ASSIGNMENT

Pardhle Ai

A Smart Study Assistant — Intelligent Agent Formulation & PEAS Evaluation

1. AI Agent Overview

Agent Name: Pardhle AI

Agent Type: Goal-Based / Learning Agent

Main Goal: Help students study smarter, stay organized, and achieve better academic performance.

Pardhle AI is an intelligent agent designed to help students manage their studies efficiently. It observes information such as deadlines, study hours, quiz results, and completed tasks through its sensors. It analyzes this information, makes decisions, and uses its actuators to create study plans, send reminders, recommend topics, and provide progress reports.

2. Graphical Representation of the Agent

The diagram below links every element of the agent into a single working cycle: sensors feed data into Pardhle AI's decision loop, the agent's decisions drive its actuators, actuators act on the environment to produce outcomes, and those outcomes feed back into the sensors — closing the learning loop.

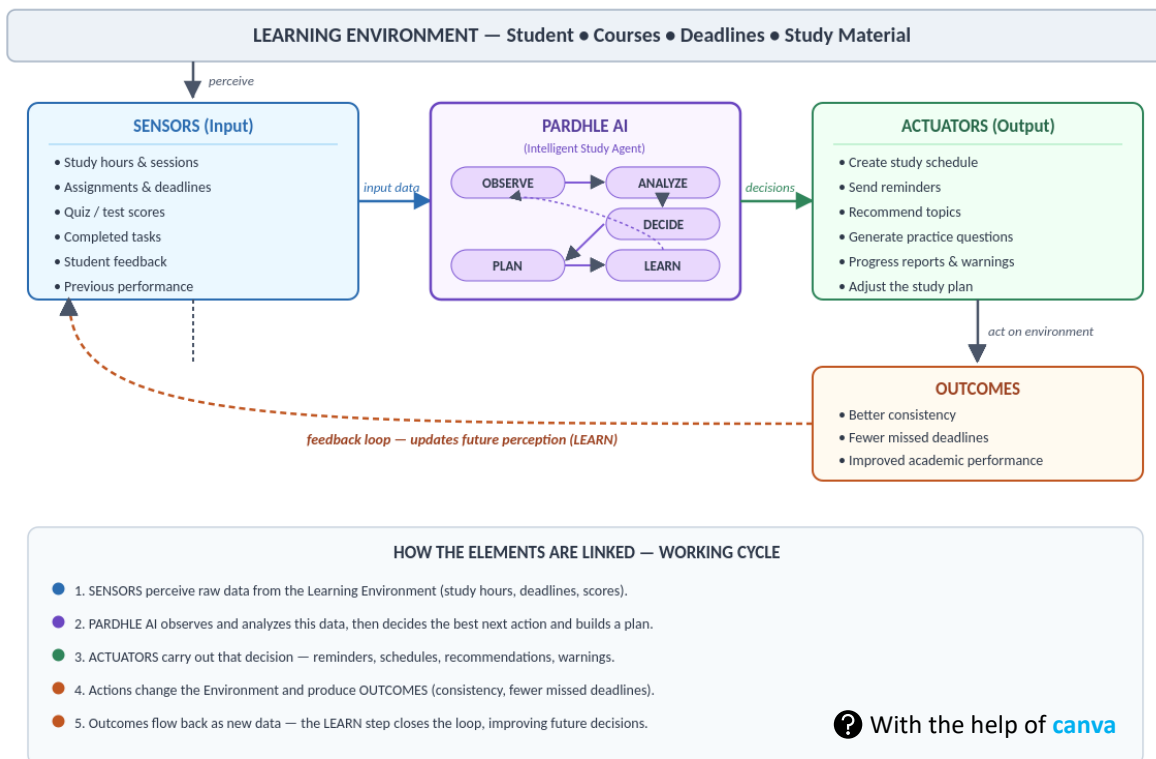


Figure 1: Pardhle AI perception–action cycle, showing how sensors, the agent, actuators, and outcomes are linked.

3. Agent Formulation

3.1 Environment

Pardhle AI operates in a digital learning environment. The environment includes the student, courses, assignments, exams, study material, calendar information, and quiz/test results.

3.2 Environment Types

Environment Property	Type	Reason
Observability	Partially Observable	The agent cannot directly observe everything about the student's understanding, mood, or motivation.
Deterministic / Stochastic	Stochastic	Student behavior and academic performance can change unexpectedly.
Episodic / Sequential	Sequential	Current study decisions can affect future performance.
Static / Dynamic	Dynamic	Deadlines, schedules, and student progress continuously change.
Discrete / Continuous	Mostly Discrete	Tasks, quizzes, deadlines, and study sessions can be represented as individual events.
Single / Multi-Agent	Single-Agent	Pardhle AI is the primary intelligent agent making decisions for the student.

3.3 Sensors

Sensors are the inputs through which Pardhle AI perceives its environment.

- Assignment information
- Calendar and deadlines
- Study time
- Quiz/test scores
- Completed tasks
- Student feedback
- Previous performance

3.4 Actuators

Actuators are the mechanisms through which Pardhle AI takes action.

- Create a study schedule
- Send reminders
- Recommend study topics
- Generate practice questions

- Display progress reports
- Warn about upcoming deadlines
- Adjust the study plan according to performance

4. PEAS Framework Evaluation

PEAS stands for Performance Measure, Environment, Actuators, and Sensors. It provides a structured way to specify what an intelligent agent should achieve, where it operates, how it acts, and what information it perceives.

PEAS Component	Pardhle AI
Performance Measure	Better grades, completed tasks, improved study consistency, fewer missed deadlines, and effective use of study time.
Environment	Student, courses, assignments, exams, calendar, study material, quizzes, and learning activities.
Actuators	Study schedule creation, reminders, topic recommendations, practice questions, notifications, warnings, and progress reports.
Sensors	Study time, deadlines, quiz scores, completed tasks, assignment information, previous performance, and student feedback.

5. Working Cycle of Pardhle AI

- Perceive — Sensors collect study-related information.
- Analyze — The agent evaluates deadlines, performance, and progress.
- Decide — The agent selects the most useful next action.
- Act — Actuators send reminders, create schedules, or provide recommendations.
- Learn — Feedback and new results are used to improve future decisions.

6. Conclusion

Pardhle AI is a simple example of an intelligent agent that applies the core concepts of Artificial Intelligence. It perceives its environment using sensors, processes information to make decisions, and performs actions through actuators. Its PEAS formulation clearly defines its performance goals, environment, available actions, and sources of information. By continuously using feedback, the agent can help a student become more organized and improve academic performance.